

## URZHAR, Kazakhstan

WMO: 366390

Lat: 47.117N

Lon: 81.617E

Elev: 1611

StdP: 13.86

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -26.9      | -21.0 | -30.6                         | 1.0 | -26.6 | -25.3 | 1.4 | -20.8 | 17.4                  | 15.7 | 12.1 | 18.1 | 0.7                      | 130  | 0.514 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 28.2                   | 94.5            | 65.7 | 91.4 | 64.6 | 88.4 | 63.9 | 68.4                | 86.0 | 67.0 | 84.4 | 65.8 | 82.9 | 3.8                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 63.2                            | 92.1 | 73.3 | 61.5 | 86.6 | 72.5 | 59.9 | 81.8 | 71.6 | 33.7          | 86.2 | 32.5 | 84.5 | 31.5 | 83.0 | 78.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 18.8          | 14.1 | 10.6 | -32.4                      | 99.5 | 6.0                | 3.1 | -36.7                                              | 101.7 | -40.2      | 103.4 | -43.6      | 105.2 | -48.0      | 107.4 |    |    |
| (5)               |      |      | -32.3                      | 71.9 | 5.9                | 2.6 | -36.6                                              | 73.8  | -40.1      | 75.4  | -43.4      | 76.8  | -47.7      | 78.7  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 42.1        | 5.7   | 11.4  | 27.0  | 48.1  | 58.9  | 68.1  | 71.5  | 68.7  | 58.2  | 45.1  | 28.9  | 11.8  |  |  |
|                                                                        | DBStd       | 24.90       | 11.60 | 10.88 | 11.29 | 9.19  | 6.91  | 5.82  | 5.05  | 5.98  | 7.70  | 8.20  | 11.69 | 12.21 |  |  |
|                                                                        | HDD50       | 5351        | 1374  | 1080  | 715   | 137   | 10    | 0     | 0     | 0     | 20    | 195   | 634   | 1186  |  |  |
|                                                                        | HDD65       | 8885        | 1839  | 1500  | 1179  | 508   | 211   | 34    | 6     | 31    | 224   | 618   | 1084  | 1651  |  |  |
|                                                                        | CDD50       | 2469        | 0     | 0     | 1     | 80    | 286   | 544   | 668   | 580   | 267   | 42    | 1     | 0     |  |  |
|                                                                        | CDD65       | 525         | 0     | 0     | 0     | 1     | 21    | 128   | 209   | 145   | 21    | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 8959        | 0     | 0     | 0     | 83    | 533   | 1956  | 2983  | 2646  | 731   | 27    | 0     | 0     |  |  |
|                                                                        | CDH80       | 3795        | 0     | 0     | 0     | 14    | 143   | 794   | 1379  | 1228  | 236   | 1     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 2.4         | 1.2   | 1.4   | 2.3   | 4.1   | 4.0   | 2.7   | 2.3   | 2.6   | 2.8   | 2.3   | 1.6   | 1.2   |  |  |
|                                                                        | PrecAvg     | 17.4        | 1.7   | 1.3   | 1.1   | 1.5   | 1.5   | 1.1   | 1.4   | 0.7   | 0.8   | 2.0   | 2.4   | 1.9   |  |  |
| (15) Precipitation                                                     | PrecMax     | 27.2        | 5.2   | 3.3   | 3.3   | 3.7   | 4.3   | 3.6   | 5.1   | 2.0   | 2.4   | 6.3   | 5.6   | 4.6   |  |  |
|                                                                        | PrecMin     | 9.6         | 0.1   | 0.1   | 0.1   | 0.0   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.3   | 0.3   | 0.2   |  |  |
|                                                                        | PrecStd     | 4.6         | 1.1   | 0.8   | 0.8   | 1.2   | 1.2   | 0.8   | 1.2   | 0.6   | 0.6   | 1.5   | 1.5   | 1.3   |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 33.4  | 38.2  | 64.3  | 81.4  | 89.0  | 94.8  | 99.1  | 97.8  | 92.2  | 77.5  | 61.4  | 38.5  |  |  |
|                                                                        |             | MCWB        | 31.1  | 32.3  | 48.7  | 59.2  | 62.3  | 66.7  | 68.3  | 66.2  | 63.9  | 56.7  | 47.8  | 32.9  |  |  |
|                                                                        | 2%          | DB          | 30.4  | 34.3  | 54.9  | 75.8  | 84.0  | 91.1  | 94.8  | 94.3  | 86.1  | 72.6  | 54.7  | 33.8  |  |  |
|                                                                        |             | MCWB        | 28.5  | 30.5  | 44.7  | 56.1  | 60.7  | 65.0  | 66.2  | 65.0  | 61.2  | 53.9  | 44.0  | 31.1  |  |  |
|                                                                        | 5%          | DB          | 26.9  | 31.6  | 47.5  | 71.3  | 79.8  | 88.2  | 91.2  | 91.0  | 82.2  | 67.7  | 48.5  | 31.2  |  |  |
|                                                                        |             | MCWB        | 24.8  | 28.7  | 40.5  | 53.7  | 58.8  | 64.2  | 65.1  | 64.0  | 59.8  | 51.8  | 41.7  | 28.8  |  |  |
|                                                                        | 10%         | DB          | 23.6  | 28.4  | 41.9  | 66.9  | 76.0  | 84.8  | 87.9  | 87.3  | 77.9  | 62.2  | 43.4  | 28.4  |  |  |
|                                                                        |             | MCWB        | 21.9  | 25.7  | 36.2  | 51.6  | 57.7  | 63.3  | 64.5  | 63.2  | 57.9  | 49.5  | 38.9  | 26.4  |  |  |
|                                                                        | 0.4%        | WB          | 32.2  | 33.5  | 50.3  | 60.9  | 65.1  | 70.1  | 71.1  | 69.2  | 65.1  | 57.7  | 49.8  | 34.0  |  |  |
|                                                                        |             | MCDB        | 33.3  | 36.3  | 61.9  | 77.2  | 83.1  | 86.8  | 89.6  | 85.7  | 87.2  | 74.6  | 57.6  | 36.6  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB          | 28.7  | 31.2  | 45.5  | 57.4  | 62.6  | 67.9  | 68.8  | 67.2  | 62.7  | 55.0  | 45.3  | 31.5  |  |  |
|                                                                        |             | MCDB        | 30.2  | 33.8  | 54.3  | 72.8  | 78.2  | 84.8  | 86.9  | 86.1  | 82.2  | 70.0  | 52.9  | 33.4  |  |  |
|                                                                        | 5%          | WB          | 25.0  | 28.7  | 40.7  | 54.8  | 60.9  | 66.3  | 67.4  | 65.9  | 60.7  | 52.7  | 42.1  | 29.0  |  |  |
|                                                                        |             | MCDB        | 26.6  | 31.4  | 47.1  | 68.2  | 74.9  | 82.1  | 84.5  | 85.2  | 79.0  | 65.3  | 47.7  | 30.8  |  |  |
|                                                                        | 10%         | WB          | 21.9  | 25.6  | 36.7  | 52.7  | 59.1  | 64.8  | 66.1  | 64.5  | 58.7  | 50.3  | 39.0  | 26.4  |  |  |
|                                                                        |             | MCDB        | 23.6  | 28.2  | 41.5  | 64.6  | 72.7  | 80.4  | 82.6  | 83.7  | 75.6  | 60.9  | 43.3  | 28.2  |  |  |
|                                                                        | 5% DB       | MDBR        | 23.5  | 26.0  | 22.4  | 25.1  | 26.8  | 27.5  | 28.2  | 31.9  | 32.0  | 27.1  | 21.5  | 22.1  |  |  |
|                                                                        |             | MCDBR       | 21.3  | 25.8  | 25.7  | 35.5  | 33.2  | 33.3  | 34.6  | 37.7  | 38.7  | 37.1  | 28.8  | 21.0  |  |  |
|                                                                        |             | MCWBR       | 18.9  | 21.5  | 16.6  | 18.3  | 14.7  | 12.9  | 12.3  | 14.2  | 17.8  | 20.5  | 18.3  | 17.4  |  |  |
|                                                                        |             | MCWBR       | 19.0  | 23.5  | 24.9  | 30.8  | 28.3  | 28.6  | 29.1  | 32.1  | 36.0  | 33.3  | 26.2  | 18.2  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.313 | 0.346 | 0.398 | 0.446 | 0.431 | 0.434 | 0.442 | 0.402 | 0.369 | 0.370 | 0.349 | 0.311 |  |  |
|                                                                        |             | taud        | 2.029 | 1.992 | 1.990 | 1.995 | 2.097 | 2.158 | 2.156 | 2.250 | 2.285 | 2.175 | 2.101 | 2.052 |  |  |
|                                                                        | Ebn at Noon |             | 230   | 249   | 255   | 255   | 265   | 264   | 260   | 266   | 265   | 240   | 215   | 214   |  |  |
|                                                                        |             | Edh at Noon | 31    | 40    | 48    | 52    | 49    | 46    | 46    | 40    | 35    | 33    | 29    | 27    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 569   | 895   | 1262  | 1660  | 2108  | 2202  | 2096  | 1938  | 1515  | 919   | 516   | 419   |  |  |
|                                                                        | RadStd      |             | 35    | 46    | 88    | 110   | 91    | 117   | 127   | 100   | 87    | 80    | 48    | 37    |  |  |

## Historical Trends

|                             |  | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|--|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |  | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |  | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           |  | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (3 neighbors) |  | N/A     | N/A | N/A    | N/A     | -0.72 | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page